

# CRUCIBLE: A Multi-Agent Framework for Automated Annotation Rubric Calibration via Iterative Inter-Judge Disagreement Resolution

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## Abstract

Annotation quality is a limiting factor in supervised learning pipelines that rely on the LLM-as-judge paradigm. This paper’s central finding is that automated inter-judge convergence is *not sufficient evidence of alignment with human judgment*—we demonstrate this four ways, on the same underlying corpus and (for three of the four) the same 30-item human reference set, so these should be read as related diagnostic analyses rather than fully independent replications: a human validation study showing one domain’s high inter-judge agreement masked a systematic divergence from human judgment; a frontier-model baseline showing a calibrated rubric does not transfer to a different judge model; a reasoning-before-label ablation showing explicit reasoning can raise inter-judge agreement without raising agreement with human judgment; and a judge-replacement test on the one local judge diagnosed as the bottleneck, which raised inter-judge agreement substantially ( $0.700 \rightarrow 0.883$ ) by converting near-unanimous dissent into full unanimity, while leaving the panel’s majority-vote output—and its accuracy against human-labeled transcripts—bit-for-bit identical: the two retained judges already agreed with each other on 90% of this domain’s transcripts, so the third judge could not have changed the majority label on this set regardless of which model it was (Section 6.3). We reach this finding while building and evaluating **CRUCIBLE**, a multi-agent system that automatically calibrates annotation rubrics through iterative disagreement diagnosis and targeted revision. Manual rubric refinement does not scale and introduces annotator-specific bias that usually goes unaudited and is hard to reproduce. Three heterogeneous local judge models independently evaluate a transcript corpus; a coordinator model (Claude Sonnet) analyzes per-model criterion-level disagreement patterns and proposes rubric revisions; the loop repeats until all domains converge or a fixed iteration budget is exhausted, flagging any domain whose agreement stagnates. CRUCIBLE was developed to solve a concrete annotation bottleneck in building a Signal-to-Noise Ratio (SNR) classifier for YouTube educational content, motivated by an earlier, differently-scoped pilot (a frontier chat-model ensemble applying a domain-agnostic rubric, reported in the companion SNR paper) that showed substantial, inconsistent inter-judge disagreement. Applied to 60 YouTube transcripts across three content domains—General Education, Technology & AI, and Career & Self-Improvement—CRUCIBLE reached 96.7% and 81.7% weighted inter-judge agreement on two domains, while the third domain failed to converge, its best rubric reaching only 70.0%, near the floor of the weighted metric. We initially read this as a pure model-capability ceiling; our later reasoning-before-label and human-validation analyses instead suggest a shared misalignment between judge models and human annotators over what

counts as “substantive” career content plausibly contributes alongside any capability limit of the judge models in isolation—though, per the judge-replacement finding above, we cannot rule out that the ceiling on this specific panel was partly a structural artifact of two judges already agreeing with each other. Because raw agreement is inflated by label prevalence, we report chance-corrected agreement (Fleiss’  $\kappa$  and Gwet’s AC1), which are less affected by the prevalence paradox under these marginals than raw agreement; these—read alongside the human validation study—locate CRUCIBLE’s value less in raising raw agreement than in diagnosing whether residual disagreement is rubric-fixable, capability-bounded, or prevalence-dominated, and in showing when that automated diagnosis itself needs an external check. We additionally check whether the calibrated rubrics’ effect on agreement generalizes to a 90-transcript corpus never seen by the coordinator during revision; the weighted-agreement metric improves there too, though the underlying majority-vote annotations themselves do not change on any of the 90 transcripts (Section 5.3; see Limitations for the precise sense in which this corpus is unseen). A pre-run discriminative-validity screening check identified a degenerate judge (Gemma 2 9B, producing identical LOW labels for all 60 transcripts) during development, before the calibration run reported here; it was replaced with Qwen 2.5 7B. The calibrated rubrics, applied by a separate frontier ensemble (reported in the companion paper), produced labels that trained a binary classifier on the SNR task achieving  $F1 = 0.871$  with  $\text{recall} = 0.964$  on the HIGH-signal class as a downstream case study; we separately find that this same frontier ensemble tracks human judgment about as loosely as the local calibration panel does. Code and calibration logs are available at <https://github.com/biditdas18/rubric-calibration-agent>.

## 1 Introduction

The LLM-as-judge paradigm has become a standard component of annotation pipelines in settings where human labeling is expensive, slow, or subject to individual bias [1]. A single LLM prompted with an evaluation rubric can produce labels at scale, but single-model annotation carries systematic biases rooted in the model’s training distribution, instruction-following style, and boundary-case interpretation [2]. Multi-model ensembles reduce this risk, but introduce a new dependency: the rubric must be precise enough that multiple independent models reach consistent conclusions. A poorly specified rubric produces disagreement not because the content is genuinely ambiguous, but because different models resolve underspecified criteria differently.

This paper originates from a concrete failure of that kind. We were building a Signal-to-Noise Ratio (SNR) framework<sup>1</sup> for binary classification of YouTube educational video transcripts—separating substantive instructional content (HIGH signal: concrete actionable steps, methodological coherence, named frameworks) from low-value filler (LOW signal: generic motivational advice, fear-based framing, promotional material, repetition without progression). The task required labeled training data reliable enough to train a lightweight classifier deployable at inference scale.

During development, an earlier annotation pipeline—a differently-scoped, differently-modeled experiment reported in the companion SNR paper [24]—showed substantial and inconsistent disagreement across judges, motivating the need for systematic rubric diagnostics.<sup>2</sup> Two remedies were available: (1) manually label a gold dataset per domain, or (2) automatically calibrate the

<sup>1</sup>The SNR classifier is described in a companion paper: Das, B. (2026). Signal-to-Noise Ratio Detection in Educational Video Content: A Binary Classification Framework Using Transcript Embeddings. SSRN. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=6866745](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6866745). That work cites CRUCIBLE as its annotation methodology.

<sup>2</sup>That pilot used a domain-agnostic rubric applied across a frontier chat-model ensemble (GPT-4, Gemini, Meta AI), distinct from the local judge panel (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B) used for calibration in this paper, and a 3-category (HIGH/MID/LOW) label space rather than the binary space CRUCIBLE’s own weighted- $A$  metric (Eq. 1) operates in. See Appendix B for the exact figures and why they are not directly comparable in magnitude to CRUCIBLE’s own convergence numbers.

rubrics until the ensemble converges. Option 1 was rejected on two grounds. First, reviewing transcripts of 10–15 minute videos at the scale required is prohibitively time-intensive. Second, and more fundamentally, manual annotation introduces annotator-specific bias that is reducible only through multiple annotators and adjudication: an annotator who values career development content intrinsically applies a systematically different threshold than one who does not. Without a multi-annotator comparison, that bias goes unaudited and undisclosed—baked into every label silently.

We designed CRUCIBLE to solve this. It is a coordinator-mediated, iterative calibration loop requiring no human reference labels and no domain expertise beyond an initial rubric draft. Three heterogeneous local models serve as judges; a frontier coordinator model diagnoses their disagreement patterns and proposes targeted rubric revisions; the loop repeats until convergence or a fixed iteration budget is exhausted. The resulting domain-specific rubrics are versioned and auditable. The labels they produce are silver quality—not gold—but empirically validated against a measurable inter-judge criterion rather than a single annotator’s unchecked judgment.

The downstream classifier serves as a case study in whether the calibrated labels are usable at scale, not as an independent validation of their agreement with the available human reference (Section 5 and Limitations address that question directly). The rubrics produced by CRUCIBLE were applied by a separate frontier ensemble (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B; reported in the companion paper) to label transcripts; an SVM trained on 389 transcripts (89 real, labeled via this calibrated-rubric ensemble methodology, plus 300 synthetically generated transcripts with labels assigned by construction, intended to match the generating prompt’s target class) achieved  $F1=0.871$  with  $\text{recall}=0.964$  on the HIGH-signal class, evaluated on a disjoint held-out set of 60 real transcripts, compared to a majority-class baseline of  $F1=0.000$  and, in a synthetic-only training ablation reported in the companion paper [24],  $F1=0.747$  when trained on the 300 synthetic transcripts alone—indicating the real, calibrated-rubric-labeled transcripts contribute materially beyond what synthetic data alone provides. We did not attempt training on labels from the uncalibrated pilot pipeline described above; the majority-class baseline ( $F1=0.000$ ) indicates the task itself is non-trivial.

Beyond the calibration loop and this downstream case study, a set of follow-up analyses—some conducted after the initial calibration run, one (judge replacement, Section 6.3) run as a direct, controlled test of this paper’s own prescriptive claim—surfaced this paper’s central finding: CRUCIBLE’s own convergence signal is not by itself sufficient evidence of label quality. This paper is organized around two questions those analyses answer: (1) does disagreement-driven rubric revision generalize past the transcripts it was tuned on, and (2) does higher inter-judge agreement correspond to higher agreement with human judgment. An external human validation study, a frontier-model rubric-transfer test, a reasoning-before-label ablation, and a direct judge-replacement test each speak to question (2); an evaluation on a 90-transcript corpus unseen by the coordinator during revision speaks to question (1). We report these alongside the calibration results because they qualify how those results should be read, and we return to both questions explicitly in Sections 5–6.

**The core contributions of this paper are:**

1. **Automated inter-judge convergence is not sufficient evidence of alignment with human judgment.** We demonstrate this four ways: a human validation study (Section 5.6) shows Technology & AI’s high inter-judge agreement (81.7%) masked a systematic divergence from human judgment; a frontier-model baseline (Section 5.8) shows the calibrated rubric does not transfer to a different judge model, instead teaching it to mimic the local ensemble’s specific

errors rather than improving its agreement with human judgment; a reasoning-before-label ablation (Section 5.8.3) shows explicit reasoning can raise inter-judge agreement without raising agreement with human judgment; and a judge-replacement test (Section 6.3) raises inter-judge agreement substantially ( $0.700 \rightarrow 0.883$ ) while leaving the panel’s majority-vote output—and its agreement with human-labeled transcripts—completely unchanged, since the two retained judges already agreed with each other on 90% of this domain’s transcripts, so the third judge could not have changed the majority label regardless of which model it was; this fourth result is narrower than the raw agreement numbers alone suggest (Section 6.3 discusses the implications). The first three share a common 30-item human-reference set and are related diagnostic analyses rather than independent replications; the fourth is a single controlled trial on the same reference set. Together they show agreement metrics can be high, or can be made to rise, while agreement with the available human reference is not.

2. **An automated rubric calibration loop** with coordinator-mediated disagreement diagnosis and targeted revision. Unlike prior rubric-generation and refinement systems, which are validated against reference judgments or accuracy signals, CRUCIBLE studies a narrower setting: iterative rubric revision driven solely by inter-judge disagreement, with no human reference labels used during calibration.
3. **A held-out generalization check.** Re-scoring both the initial and final calibrated rubrics on a 90-transcript corpus never shown to the coordinator during revision (Section 5.3) shows the revision’s effect on the weighted inter-judge *agreement metric* generalizes past the calibration corpus for the two domains that were actually revised, with a positive, CI-excluding-zero effect for Career & Self-Improvement specifically—though, as detailed there, this generalization is driven disproportionately by one judge’s behavior shifting, not uniform three-way convergence, and the underlying majority-vote annotations do not change on any of the 90 transcripts under either rubric.
4. **The capability-ceiling finding, revised.** The system empirically distinguishes rubric-fixable disagreement (Technology & AI:  $76.7\% \rightarrow 81.7\%$  in one iteration) from disagreement that further rubric revision did not resolve (Career & Self-Improvement: failed to converge despite two revisions, with its best rubric at 70.0%). We initially attributed this to a pure model-capability limitation; the reasoning-before-label ablation and human validation study (Sections 5.8.3 and 5.6) suggest the ceiling may instead partly reflect a shared misalignment between judge models and human annotators over what counts as “substantive” career content, though we cannot fully separate this from a structural property of the panel itself: our judge-replacement test (Section 6.3) found the two static judges already agreed with each other on 90% of this domain’s transcripts, so no third judge could have changed the panel’s majority-vote output on this set regardless of the ceiling’s underlying cause.
5. **Per-domain convergence.** Domains converge independently, each against its own rubric and disagreement history. Two supporting quality-control components are detailed in Sections 3.5–3.6: a discriminative-validity pre-screening gate that caught and removed a degenerate judge (Gemma 2 9B, zero discriminative outputs) before the reported run, and a columnar evaluation order that made repeated 60-transcript, 3-model passes tractable on consumer hardware. Each iteration also re-evaluates any domain that converged on the immediately preceding iteration, as a one-time check that its agreement rate holds on a second pass with the same rubric and judges (Section 6.5)—an output-stability check, not a cross-domain interference test, since coordinator calls are single-domain, stateless API requests with no mechanism for one domain’s revision to affect another’s.

6. **A downstream case study.** The calibrated rubrics, applied by a separate frontier ensemble, produced labels that trained an SVM classifier achieving  $F1=0.871$  and  $\text{recall}=0.964$  on the HIGH-signal class (Section 5). We treat this as a usability case study, not an independent validation of the labels’ agreement with the available human reference: Section 5.7 shows the same frontier ensemble tracks human judgment about as loosely as the local calibration panel does.

## 2 Related Work

### 2.1 LLM-as-Judge

The use of large language models as automated evaluators was formalized by Zheng et al. [1] in the MT-Bench benchmark. Subsequent work extended this paradigm to structured rubrics [3], fine-grained multi-dimensional scoring [4], and customizable evaluation dimensions [5]. A persistent finding is that single-model judges exhibit systematic biases—position bias, verbosity bias, and self-preference—that reduce reliability [1, 2]. Multi-model ensembles have been proposed to mitigate these biases [6], but ensemble reliability still depends on rubric quality.

### 2.2 Rubric Design and Calibration

LLM-Rubric [7] combines LLM predictions with a small neural network to predict human judge annotations. AdaRubric [8] proposes task-adaptive rubric generation for agent evaluation. Autorubric [9] provides a unifying framework for rubric-based evaluation across criterion types. Prior rubric-refinement work that does iterate does so against human-labeled reference data [10]. Our narrower claim relative to this body of work is not that automated rubric revision itself is unaddressed in general, but that we revise domain-specific annotation rubrics using cross-model disagreement as the sole feedback signal, with no human reference labels used during calibration, while explicitly investigating when that signal produces panel-specific consensus rather than alignment with human judgment (Sections 5.6–5.8.3) — a question the reference-judgment-validated systems above are not positioned to ask, since their validation signal is human-labeled by construction. SciIF [11] pairs problems with a fixed catalog of scientific constraints and structured verification rubrics, but treats the rubric as fixed rather than something to be revised. Recent work also automates rubric generation and refinement: RRD [17] recursively decomposes and filters rubric criteria, and GER-Eval [18] has LLMs design and apply their own rubrics. Both are still validated against reference judgments or accuracy signals; CRUCIBLE instead drives revision purely from inter-judge agreement with no human reference labels, and separately audits, via human validation, when that agreement-only signal is and is not trustworthy.

### 2.3 Disagreement Diagnosis

Related self-deliberation mechanisms appear in evaluation frameworks such as AraTable [12], where a judge re-examines its own assessment. Multi-agent debate [13] resolves disagreement through structured exchange among models. These approaches are self-revision or debate; neither produces a generalized rubric revision that improves agreement across the full corpus and on future evaluation runs.

## 2.4 Multi-Agent Coordination

Multi-agent debate frameworks [13] have been applied to improve LLM output reliability through structured disagreement resolution. CRUCIBLE applies the coordinator-mediator pattern asymmetrically: judge models do not communicate with each other; only the coordinator sees aggregate disagreement patterns and produces the revision.

## 2.5 Annotation Bias and Scalability

The annotation bottleneck has been addressed through weak supervision [14], programmatic labeling [15], and knowledge distillation from frontier models [16]. CRUCIBLE is complementary: it produces rubrics calibrated against inter-judge agreement, grounding ensemble annotation at scale—though, as Section 5.6 shows, that calibration signal alone does not guarantee the rubrics are human-validated.

# 3 The CRUCIBLE System

## 3.1 Problem Statement

Given a corpus  $\mathcal{D} = \{d_1, \dots, d_N\}$  to be annotated with a binary label  $y \in \{H, L\}$ , and an initial rubric  $R^{(0)}$  specifying criteria for each label, we seek a revised rubric  $R^{(k)}$  such that the weighted inter-judge agreement  $A(R^{(k)}, \mathcal{D}) \geq \tau$  for a target threshold  $\tau$ , with the revision process requiring no human-labeled reference data.

The key insight is that inter-judge disagreement encodes rubric underspecification: when two competent models applying the same rubric reach opposite conclusions on the same document, the rubric has not resolved the case. A coordinator with visibility into both decisions and the criterion-level justifications can identify *which criterion was misapplied and how*—and propose a targeted revision to close the gap.

## 3.2 Architecture

Figure 1 illustrates the full system. Judge models operate independently and in isolation from each other; only the coordinator sees their combined outputs. The rubric revision loop is the only feedback path—there is no human in the loop after initialization.

## 3.3 System Components

**Judge models.** Three heterogeneous local models running at temperature 0.0 via Ollama: Llama 3.1 8B (Meta), Mistral 7B (Mistral AI), and Qwen 2.5 7B (Alibaba). Temperature 0.0 yields near-deterministic outputs, isolating rubric quality as the variable of interest. The three models were chosen to represent different training lineages, maximizing the diagnostic value of their disagreements. These local models serve as the calibration instrument only; production labeling in the downstream application uses a separate, stronger frontier ensemble (companion paper).

**Coordinator model.** Claude Sonnet (claude-sonnet-4-6) [22], via API. The coordinator receives: (a) the current rubric, (b) per-model label assignments and criterion-level justification quotes for each disagreement case, and (c) disagreement counts per criterion. It produces a structured JSON revision proposal specifying which criterion to modify and the revised text.



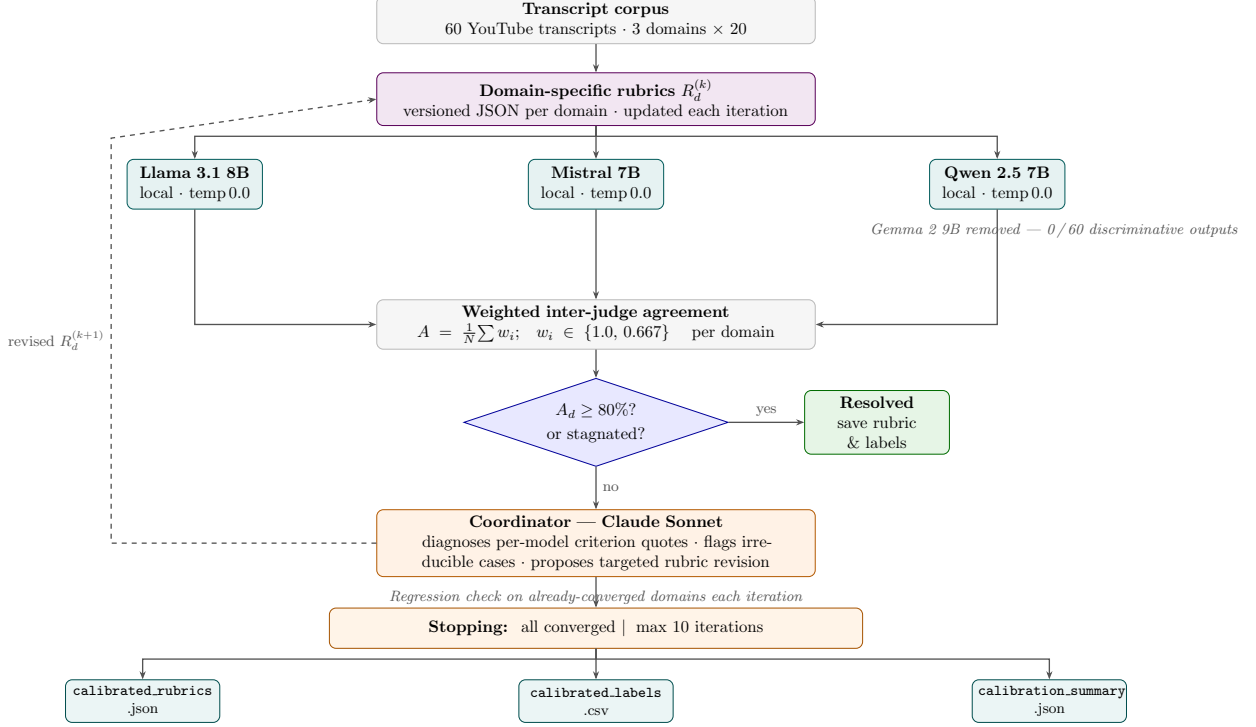


Figure 1: CRUCIBLE system architecture. Judge models operate independently at temperature 0.0. The dashed arrow is the only feedback path: revised rubrics flow back into the next iteration. No human input occurs after initialization.

### Agreement metric.

$$A = \frac{1}{N} \sum_{i=1}^N w_i, \quad w_i = \begin{cases} 1.0 & \text{all three judges agree} \\ 0.667 & \text{exactly two judges agree} \end{cases} \quad (1)$$

For three judges on a binary label, every item is either unanimous ( $w = 1.0$ ) or a two-one majority ( $w = 0.667$ ); the no-majority case cannot occur, so  $A$  is bounded below by 0.667. Under independent judges the expected value of  $A$  is approximately 0.75, which should be read as the chance level for this metric; label skew can raise it further.

All-agree rate (unanimous three-way agreement) is tracked separately as a stricter secondary metric.

### 3.4 Calibration Loop

**Initialization.** Domain-specific rubrics  $R_d^{(0)}$  are written at moderate abstraction—sufficient to convey intent but not yet precise enough to resolve edge cases consistently.

**Iteration  $k$ :**

1. All three judges evaluate all documents using  $\{R_d^{(k)}\}$ .
2. Weighted agreement  $A_d^{(k)}$  is computed per domain.
3. For each non-converged, non-stagnated domain with  $A_d^{(k)} < \tau$ :
  - a. Extract disagreement cases (at least one judge diverged).
  - b. Assemble per-model criterion quotes for each case.

- c. Coordinator diagnoses and proposes revision.
  - d. If valid JSON:  $R_d^{(k+1)} \leftarrow \text{revise}(R_d^{(k)})$ .
  - e. Irreducible cases (genuine ambiguity, not rubric failure) are logged and excluded from stagnation counting.
4. Regression check on each already-converged domain.

**Stopping.** A domain is marked *stagnated* when  $\Delta A_d \leq 0$  for three consecutive iterations; it is then excluded from further revision and held at its current rubric, but this does not terminate the loop. The loop terminates only when (a) all domains have converged or (b) the maximum of 10 iterations is reached. Because Career & Self-Improvement never converged, the present run continued to the 10-iteration cap, with converged domains monitored by regression checks and the stagnated domain held in place.

### 3.5 Judge Quality Control

Prior to the primary run, each candidate judge is evaluated for discriminative validity. A degenerate judge—one assigning a single label to all documents—provides zero inter-judge signal and pollutes agreement scores.

During development, Gemma 2 9B produced LOW labels for all 60 transcripts across all domains (0% HIGH rate). This is consistent with treating LOW criteria as a checklist to confirm rather than a comparative threshold to apply. Gemma 2 9B was replaced with Qwen 2.5 7B before the calibration run reported here.

### 3.6 Columnar Evaluation Order: An Engineering Optimization

The order in which judge models are invoked across the corpus has a dramatic effect on runtime: batching evaluation by model rather than by document (running each judge across all 60 transcripts before switching models) reduces model-load events from 180 to 3 per iteration and cuts per-iteration runtime from an incomplete >30 minutes to 17–20 minutes on consumer hardware, making the full 10-iteration run tractable. This is an engineering detail, not a scientific finding; the full derivation and the naive row-wise alternative it replaces are given in Appendix C for readers interested in reproducing the setup.

## 4 Experimental Setup

**Task and corpus.** The target task is binary classification of YouTube educational video transcripts under the SNR taxonomy: HIGH signal (substantive instructional content: concrete actionable steps, methodological coherence, named frameworks) vs. LOW signal (generic motivational advice, fear-based framing, promotional material, repetition without progression). The corpus comprises 60 transcripts across three domains (20 per domain): General Education, Technology & AI, and Career & Self-Improvement.

**Sampling protocol.** Transcripts were collected via two methods: approximately 20 via manual search, and the remaining approximately 130 (spanning both this 60-transcript evaluation corpus and the 90-transcript training-supplement corpus used in Section 5.3) via the YouTube Data API v3, using a VPN to work around geographic and rate-limit restrictions. API search was constrained to medium-length videos (4–20 minutes) with closed captions available, filtered for English relevance, using domain-specific query sets (5 base queries per domain, expanded with “tips”/“tutorial” variants when needed to fill the candidate pool); candidates were deduplicated against the existing



evaluation set and filtered to 300–8000 words per transcript. Candidate selection prioritized a fixed allow-list of known channels per domain (established educational/technical channels expected to produce clear HIGH signal) before falling back to general search results—a deliberate design choice trading naturalistic diversity for more consistent signal quality, and a limitation on how representative the corpus is of YouTube educational content generally (see Limitations). This allow-list bias plausibly contributes to the strong HIGH-signal skew reported below (**Corpus label distribution**): a channel allow-list built to surface clear HIGH-signal content is, by construction, unlikely to also surface a representative share of genuinely LOW-signal content, so the corpus is better described as a curated educational-content sample than as representative of YouTube educational content generally.

**Corpus label distribution.** Both corpora are strongly HIGH-skewed at the majority-vote level. On the 60-transcript evaluation corpus (final calibrated labels), majority-vote HIGH rate is 100% for General Education, 60% for Technology & AI, and 90% for Career & Self-Improvement; the individual judges are more HIGH-skewed still (Llama 3.1 8B: 100%, 85%, 100% HIGH by domain; Mistral 7B: 100%, 60%, 90%; Qwen 2.5 7B, the more LOW-leaning judge: 90%, 30%, 10%). On the 90-transcript held-out corpus (final rubric), majority-vote HIGH rate is 100% in every domain (Llama and Mistral predict HIGH on essentially every transcript in both corpora; Qwen 2.5 7B is again the only judge with substantial LOW-rate mass: 3%, 37%, 63% HIGH by domain under the final rubric). We report these figures because they bear directly on how several results in this paper should be read—most importantly Section 6.3’s judge-replacement test, where Llama and Mistral’s near-total mutual agreement on Career & Self-Improvement leaves little room for a third judge to influence the panel’s majority-vote output (Section 6.3 quantifies this directly).

**Channel coverage.** The 60-transcript evaluation corpus spans 46 unique channels (up to 5 transcripts from a single channel); the 90-transcript training-supplement corpus (Section 5.3) spans 50 unique channels (up to 7 from a single channel). The two corpora share 6 channels in common. Consequently, “held-out” in Section 5.3 means *document*-held-out (no transcript appears in both corpora), not *source*-held-out—see Limitations.

**Initial rubrics.** Three domain-specific rubrics at version 0, each specifying HIGH-signal criteria (methodological coherence, specificity, informational novelty, solution orientation) and LOW-signal patterns (generic advice, repetition, fear-based framing, promotional content, logical fallacies) with domain-appropriate threshold language.

**Target threshold.**  $\tau = 0.80$ . This is a modest margin above the metric’s chance level of  $\approx 0.75$  under independent judges (Eq. 1), not a claim that 0.80 represents strong agreement. This  $\approx 0.75$  chance level is derived entirely from CRUCIBLE’s own binary weighted- $A$  metric and local judge panel; it is unrelated to, and should not be blended with, the separate 3-category pilot discussed in the Introduction and Appendix B, which concerns a different experiment in a different label space. We treat  $\tau$  as a practical stopping rule for the calibration loop—a minimum bar past which further iteration is not obviously worthwhile—rather than as the paper’s primary quality signal; the chance-corrected metrics (Fleiss’  $\kappa$ , Gwet’s AC1) reported in Section 5 are what we rely on to judge whether agreement reflects genuine convergence.

**Coordinator retry.** Up to 3 retries per call on JSON parsing failure; 5-second delay between attempts.

**Runtime.** 17–20 minutes per iteration (columnar order). 170.7 minutes total; the run terminated at the maximum-iteration cap of 10 iterations. Consumer MacBook (Apple Silicon), no discrete GPU; exact CPU/GPU offload configuration for the Ollama-served judge models was not recorded,

and Ollama’s default behavior on Apple Silicon is to use Metal GPU acceleration where available, so “CPU-only” should not be assumed.

**Baseline.** CRUCIBLE’s own domain-specific v0 rubrics are the calibration starting point (Table 1). The earlier, differently-scoped pilot mentioned in the Introduction (frontier chat-model ensemble, domain-agnostic rubric, 3-category label space) motivated this work but is not a directly comparable quantitative baseline for the figures below; see Appendix B.

## 5 Results

### 5.1 Per-Domain Convergence

Table 1: Calibration results per domain.

| Domain              | Initial $A$ | Best $A$ | All-agree | Fleiss $\kappa$ | Gwet AC1 | Rubric ver.    | Status             |
|---------------------|-------------|----------|-----------|-----------------|----------|----------------|--------------------|
| General Education   | 96.7%       | 96.7%    | 90.0%     | −0.034          | +0.929   | v0 (0 changes) | Converged, iter. 1 |
| Technology & AI     | 76.7%       | 81.7%    | 45.0%     | +0.246          | +0.286   | v1 (1 change)  | Converged, iter. 2 |
| Career & Self-Impr. | 68.4%       | 70.0%    | 10.0%     | −0.350          | −0.080   | v1 (best of 2) | Stagnated          |

General Education converged immediately in iteration 1 with no rubric revision. Educational content with clear structural markers (lecture format, sequential explanation, named frameworks) produced consistent judgments across all models.

Technology & AI required one revision. The coordinator diagnosed that Qwen 2.5 7B was reading HIGH criterion 1 too narrowly—requiring production-system experience where the rubric intended methodological grounding. The revised criterion made the threshold explicit; agreement rose from 76.7% to 81.7%, crossing the 80% target in one step.

Career & Self-Improvement required two revision attempts: the first (v1) reached the best agreement of 70.0%, the second (v2) regressed to 68.4% and was discarded, so v1 was retained. Agreement never approached the 80% target. The coordinator’s diagnosis across iterations 1 and 2 consistently identified the same failure: Qwen 2.5 7B applied LOW criterion 1 (generic motivational language) to any transcript containing motivational phrasing, regardless of whether procedural content co-occurred. Revisions made the co-occurrence condition progressively more explicit, but the model continued misapplying the criterion. The coordinator flagged this as a model capability ceiling: the model was failing to apply the co-occurrence condition consistently across long transcripts—a limitation at the model level, not a rubric ambiguity.

We report Fleiss’  $\kappa$  and Gwet’s AC1 alongside raw weighted agreement, because raw agreement is inflated by label prevalence. The three domains separate cleanly under chance correction. Technology & AI shows fair agreement ( $\kappa = +0.25$ ,  $AC1 = +0.286$ )—the only domain where a rubric revision carried agreement to the convergence target. Career & Self-Improvement shows below-chance agreement ( $\kappa = -0.35$ ,  $AC1 = -0.080$ ): the judges are systematically anti-correlated relative to that metric’s marginal-based chance expectation, which no rubric revision was able to change—consistent with, though not independent proof of, the capability-ceiling diagnosis discussed further in Section 6.3. General Education shows  $\kappa \approx 0$  despite 96.7% raw agreement; this is the known prevalence paradox of  $\kappa$  under skewed marginals [19] (the domain is roughly 97% one class), and Gwet’s  $AC1 = +0.929$  is less affected by the prevalence paradox under these marginals than raw agreement or Fleiss’  $\kappa$ . The practical point is that CRUCIBLE’s contribution is not raising raw agreement per se—raw agreement is base-rate-dependent—but diagnosing which regime a domain

occupies: rubric-fixable, capability-bounded, or prevalence-dominated.

**Initialization vs. iteration.** We separate how much each domain’s result owes to writing a domain-specific starting rubric (v0, iteration 1) versus the iterative revision loop that follows.<sup>3</sup> In General Education, initialization alone (iteration 1) achieved 96.7%—identical to the final result; the revision loop contributed nothing because none was needed. In Technology & AI, the domain-specific v0 rubric started at 76.7%, and one revision then added 5.0 points to cross threshold at 81.7%. In Career & Self-Improvement, the domain-specific v0 rubric started at 68.4%; one revision (v1) added a further 1.6 points to 70.0% before a second revision (v2) regressed performance and was discarded. Across all three domains, the domain-specific v0 rubric alone accounts for most of each domain’s final agreement level; the iterative loop’s marginal contribution is small in absolute terms, necessary to cross threshold in Technology & AI, and attempted-but-unhelpful in Career & Self-Improvement—itself informative, as it is what let us distinguish a rubric-fixable domain from one exhibiting the capability-ceiling pattern investigated further in Section 5.8.3 and directly tested via judge replacement in Section 6.3.

## 5.2 Convergence Trajectory

Table 2: Weighted agreement (%) per domain across iterations. ✓ = converged; △ = no improvement over best. Career’s second revision (v2) regressed at iteration 3 and was not retained; iterations 4–10 show no further improvement. The retained output is the best rubric, v1 (70.0%, iteration 2).

| Iter. | Career & SI | Tech & AI | Gen. Edu. |
|-------|-------------|-----------|-----------|
| 1     | 68.4        | 76.7      | 96.7 ✓    |
| 2     | 70.0        | 81.7 ✓    | 96.7 ✓    |
| 3     | 68.4 △      | 81.7 ✓    | 96.7 ✓    |
| 4–10  | 68.4 △      | 81.7 ✓    | 96.7 ✓    |

Both converged domains remained stable across all subsequent iterations; regression checks confirmed no drift. For Career & Self-Improvement, the second revision (v2) regressed to 68.4% at iteration 3 and was rejected; no later revision exceeded the best observed rubric, so the retained system output is v1 at 70.0% from iteration 2. Agreement never reached the 80% target.

## 5.3 Held-Out Rubric Evaluation

Table 1 and Table 2 report agreement on the same 60-transcript corpus the coordinator revised rubrics against—so they cannot by themselves distinguish genuine generalization from fitting to that specific corpus. We address this with a single inference pass (no further coordinator revision) of the v0 and final calibrated rubrics over a 90-transcript corpus (30/domain) drawn from the training-supplement collection described in Section 4. We call this the *unseen-to-coordinator* set rather than a “held-out test set”: the strongest claim we can make with confidence, based on the documented project workflow rather than a logged, timestamped audit trail, is that this corpus was not shown to the coordinator during rubric revision. As established in Section 4, it is document-

<sup>3</sup>Earlier drafts of this analysis expressed iteration 1’s contribution as a percentage of a gain measured from a 53% “domain-agnostic baseline.” That baseline comes from a different experiment (a frontier chat-model ensemble on a 3-category label space; see Appendix B) and is not in the same units as the local-panel, binary weighted-*A* figures below, so we no longer report a percentage-of-gain figure against it.

disjoint from the calibration corpus but shares 6 channels with it, so this is not a source-held-out test (see Limitations).

Table 3: Unseen-to-coordinator evaluation ( $n=30/\text{domain}$ ): weighted agreement under the v0 vs. final calibrated rubric, single inference pass, no further revision. CI is a 95% paired bootstrap interval (10,000 resamples) on the per-transcript weight difference (final minus v0).

| Domain              | v0    | Final | Diff.    | 95% CI        |
|---------------------|-------|-------|----------|---------------|
| Career & Self-Impr. | 72.3% | 78.9% | +6.7 pts | [+2.2, +11.1] |
| Technology & AI     | 84.5% | 87.8% | +3.3 pts | [+0.0, +6.7]  |
| General Education   | 98.9% | 98.9% | +0.0 pts | [+0.0, +0.0]  |

This shows the revision’s effect on inter-judge agreement is not confined to the calibration corpus. Career & Self-Improvement’s interval excludes zero; Technology & AI’s is borderline (lower bound at exactly 0); General Education’s rubric was never revised, so no change is possible or observed by construction. We report this as evidence the revision generalizes in its ability to *change inter-judge agreement* on unseen transcripts. It does **not** establish that the final rubric is more accurate or more human-aligned on this set—no human labels exist for it—and we do not describe this result as held-out validation of label quality anywhere in this paper.

The chance-corrected picture qualifies this further. Fleiss’  $\kappa$  stays negative on this held-out set for both revised domains (Career:  $-0.385 \rightarrow -0.268$ ; Technology & AI:  $-0.184 \rightarrow -0.139$ ), the same prevalence-paradox pattern documented in Section 5 for the calibration corpus; Gwet’s AC1 is the metric that actually moves (Career:  $0.072 \rightarrow 0.367$ ; Technology & AI:  $0.578 \rightarrow 0.689$ ) and is the more informative read here given  $\kappa$ ’s known instability under this label distribution. Per-judge label prevalence explains why: on this corpus, Llama 3.1 8B and Mistral 7B predict HIGH on essentially 100% of transcripts in both domains, under *both* rubric versions—they do not move. Qwen 2.5 7B is the only judge whose behavior shifts (Career:  $17\% \rightarrow 37\%$  HIGH rate; Technology & AI:  $53\% \rightarrow 63\%$  HIGH rate), and this shift is what drives the reported agreement improvement. This is Qwen’s HIGH-rate converging toward the other two, effectively static, judges—not three-way convergence toward more nuanced shared judgment. Because Llama and Mistral already agree with each other on nearly every transcript in both domains, their majority vote is mechanically fixed regardless of Qwen’s individual prediction: the revision shifted the dissenting judge’s individual predictions but changed the ensemble’s *majority-vote annotation* on **0 of the 90** held-out transcripts (0 of 30 in each domain) between the v0 and final rubric. The weighted-agreement improvement reported above reflects Qwen’s predictions moving closer to unanimity with an already-fixed majority, not a change in which label the panel actually output on any transcript. Full per-judge prevalence and provenance detail are in Appendix D.

#### 5.4 Downstream Validation: SNR Classifier Performance

We report this as a downstream *case study* in whether the calibrated labels are usable at scale, not as an independent validation of their agreement with the available human reference. The calibrated rubrics were used to drive labeling for a lightweight SVM classifier on the SNR binary classification task. Real transcripts labeled using CRUCIBLE-calibrated rubrics (applied by a separate frontier ensemble, per the companion paper [24]) were combined with 300 synthetically generated transcripts for class balance, and the classifier was evaluated on a held-out set of 60 real transcripts.

| Model                                                  | F1 (HIGH signal) | Recall (HIGH signal) |
|--------------------------------------------------------|------------------|----------------------|
| SVM on rubric-calibrated labels (mixed real+synthetic) | 0.871            | 0.964                |
| SVM on synthetic-only training (companion paper)       | 0.747            | —                    |
| Majority-class baseline                                | 0.000            | —                    |

We did not train a classifier on labels from the uncalibrated pilot pipeline (Introduction); the majority-class baseline ( $F1 = 0.000$ ) indicates the task is non-trivial. The gap between the synthetic-only ablation ( $F1 = 0.747$ , reported in the companion paper [24]) and the mixed real+synthetic result ( $F1 = 0.871$ ) indicates the real, calibrated-rubric-labeled transcripts contribute materially beyond what synthetic data alone provides—though, per Limitations, this does not fully rule out the mixed result partly reflecting the classifier learning to separate synthetic-generation artifacts rather than purely calibrated-label quality. Because the held-out labels were produced by the same silver-labeling methodology rather than independent human-validated labels, this F1 reflects internal consistency with the labeling pipeline rather than external, human-validated accuracy—Section 5.7 addresses this directly. Full experimental details of the SNR classifier—dataset construction, synthetic augmentation strategy, and complete evaluation protocol—are reported in the companion paper [24] (see footnote 1).

**Train/test split and label independence.** The classifier is trained on 389 transcripts: 89 real transcripts with silver labels assigned via the same calibrated-rubric ensemble methodology, combined with 300 synthetic transcripts (150 HIGH, 150 LOW, balanced across domains) generated from structured prompts with labels assigned by construction, intended to match the generating prompt’s target class. Evaluation uses a disjoint, held-out set of 60 real transcripts—the same 60-transcript corpus used throughout this paper—selected and labeled prior to any classifier training. The 89 training-supplement transcripts and 300 synthetic transcripts are disjoint from the 60-transcript evaluation set by construction (no transcript appears in both); synthetic transcripts correspond to no real video, and the 89 real training-supplement transcripts are document-disjoint from evaluation but—per Section 4—share some channels with it (6 of 46 evaluation-set channels also appear in the 90-transcript training-supplement pool from which the 89 are drawn), so this is document-held-out, not source-held-out. We note directly, per the concern motivating Section 5.6, that the evaluation labels are *not* independent of the calibration process in the sense of an external human reference—they are assigned by the same class of frontier-ensemble judges used for the training supplement, applying the calibrated rubric. This is precisely the circularity Section 5.6 is designed to address: human consensus and calibrated-panel labels agree at 90% (General Education), 50% (Technology & AI), and 50% (Career & Self-Improvement)—figures that should be read alongside, and as a caveat on, the F1 reported above. The F1 score demonstrates the calibrated labels are *learnable* by a downstream classifier; it does not by itself demonstrate the labels are *correct*.

## 5.5 Irreducible Cases

The coordinator flagged two Technology & AI transcripts as irreducible—genuine content ambiguity rather than rubric failures—and excluded them from stagnation counting. We were unable to independently verify these specific cases against surviving logs (Appendix E explains why and reports a sensitivity check instead), so we do not name the specific transcripts here as the original draft of this paper did.

## 5.6 Human Validation

A known limitation of LLM-as-judge pipelines is that inter-judge agreement among automated annotators may reflect shared model bias rather than agreement with independent human judgment, and without a human reference point, convergence cannot be distinguished from consensus error. We address this directly with an independent human validation study.

### 5.6.1 Design

We recruited three independent annotators per domain (nine total) via Prolific [21], stratified by domain (10 transcripts per domain, sampled from the 60-transcript corpus, plus one hidden attention-check item per domain not included in the analysis below). Annotators were required to be proficient in English and hold a Prolific approval rating above 95%. Annotators were shown only the raw transcript text—not the calibrated rubric, not the domain label, and not CRUCIBLE’s output—and asked to judge, in their own words, whether each transcript was substantive and valuable or generic and low-value, with a mandatory written justification. This is an unaided-judgment design only; we did not run a second condition showing annotators the calibrated rubric (see Limitations for what this does and does not establish). This design is deliberate: showing annotators the calibrated rubric would test whether humans can *follow* CRUCIBLE’s criteria, not whether CRUCIBLE’s criteria independently match human judgment. Quality was enforced via a hidden attention-check item per domain (adversarial or unambiguous content with a known expected label), a mandatory content-grounded justification, and task instructions designed to flag fast or low-effort responses for rejection. One of nine submissions (Technology & AI domain) was rejected and replaced, on three converging factors confirmed by reviewing the full submission: a completion time of 39:06 against this batch’s displayed median of 140:00, failure of the domain’s attention-check item, and justifications consisting largely of copy-pasted transcript excerpts rather than substantive reasoning. This was a multi-factor judgment made after reviewing the submission’s full content, not a blind, mechanically-triggered rejection; we report a sensitivity check in Appendix F showing the domain-level results are unchanged whether this submission or its replacement is used.

### 5.6.2 Results

We report two quantities per domain: (1) inter-annotator agreement among the three humans, via Fleiss’  $\kappa$  and Gwet’s AC1 [19], and (2) the agreement between human majority-vote consensus and CRUCIBLE’s calibrated label for the same transcript, with 95% bootstrap confidence intervals (10,000 resamples) given the small per-domain sample. With  $n = 10$  per domain, percentile bootstrap intervals of this kind may not fully capture the true sampling uncertainty; the intervals below should be read as indicative of the scale of uncertainty, not as precise bounds. Channel identity is recorded for this corpus (Section 4); Appendix G reports these same intervals resampled by channel rather than by transcript, since several channels contribute more than one transcript to this 30-item sample.

General Education and Technology & AI show identical  $\kappa$ /AC1 by coincidence of vote-count distribution shape across their 10 items, verified directly against the raw annotation tables, not a computation error.

The confidence intervals for Technology & AI and Career & Self-Improvement overlap substantially, and both differ from General Education’s interval only at the boundary; we report the domain-level distinction below as supported by the *pattern* across human-human and human-model agreement jointly, not by the point estimates in isolation, which alone are not statistically distinguishable at



Table 4: Human validation results per domain. Human  $\kappa$ /AC1 measure agreement among the three annotators; Human vs. CRUCIBLE reports the percentage of transcripts where human majority-vote consensus matches CRUCIBLE’s calibrated label, with 95% bootstrap CI.

| Domain              | Human $\kappa$ | Human AC1 | Human vs. CRUCIBLE | 95% CI                |
|---------------------|----------------|-----------|--------------------|-----------------------|
| General Education   | 0.04           | 0.63      | 90.0%              | [70.0%, 100.0%]       |
| Technology & AI     | 0.04           | 0.63      | 50.0%              | [20.0%, 80.0%]        |
| Career & Self-Impr. | 0.33           | 0.33      | 50.0%              | [20.0%, 80.0%]        |
| <b>Overall</b>      | —              | —         | <b>63.3%</b>       | <b>[46.7%, 80.0%]</b> |

this sample size.

These results support three distinct conclusions, one per domain—with one important caveat throughout: each domain was validated by a distinct, non-overlapping set of three annotators (nine total, no annotator contributed to more than one domain), so domain effects and annotator-cohort effects cannot be fully separated with this design. Each conclusion below should be read as characterizing the outcome for that domain’s specific three-person annotator group, which we cannot cleanly distinguish from a property of the domain’s content itself; a design with overlapping or larger annotator pools per domain would be needed to separate these two sources of variance.

**General Education: this domain’s annotator group validated CRUCIBLE’s labels.**

These annotators agree with each other at a moderate-to-good level (AC1 = 0.63) and agree with CRUCIBLE’s label 90% of the time. This is consistent with this domain’s fast, stable convergence during calibration (Table 1: 96.7% agreement, v0 rubric, no revisions) and is consistent with convergence here reflecting genuine label validity, not shared model artifact, for this annotator group.

**Technology & AI: this domain’s annotator group’s judgments diverged from CRUCIBLE despite agreeing with each other.**

These annotators agree with each other at the same rate as General Education’s annotators (AC1 = 0.63)—for this group, the task was well-posed and answerable consistently. Yet their consensus matches CRUCIBLE’s label only 50% of the time. Because human-human agreement is high while human-CRUCIBLE agreement is low, this is not consistent with an inherently ambiguous task for this annotator group; it is evidence that CRUCIBLE’s confident convergence (81.7% inter-judge agreement, crossing our calibration threshold) *masked* a systematic divergence from this group’s judgment—a known failure mode in LLM-as-judge pipelines: high automated agreement gives no signal that labels are unreliable when the automated panel and the human reference diverge in a way agreement metrics alone cannot detect, and Technology & AI’s agreement statistics here are empirically indistinguishable from General Education’s. These annotators were screened for English proficiency and Prolific approval rating, not for technical or AI/ML domain expertise, which may itself contribute to the divergence in ways we cannot separate from a genuine CRUCIBLE misalignment with this group’s judgment.

**Career & Self-Improvement: this domain’s annotator group itself showed low agreement.**

These annotators show markedly lower agreement with each other (AC1 = 0.33) than the other two groups. Since this group could not consistently agree among themselves, CRUCIBLE’s own stagnation here (Table 1: 68.4%–70.0%, never crossing threshold) is at least partly attributable to genuine task difficulty for this group rather than model failure alone. This nuance—without fully resolving—our capability-ceiling diagnosis (Section 5.8.3 tests this directly): the ceiling may be substantially a property of the content itself as judged by this group, not solely the judge models.

### 5.6.3 Implications

These results require us to revise the framing of CRUCIBLE’s central claim. Automated inter-judge convergence is *not sufficient evidence of alignment with human judgment*: it correctly identified General Education’s annotator group as reliable and correctly flagged Career & Self-Improvement’s as difficult, but it failed to distinguish Technology & AI’s unreliable convergence from General Education’s valid convergence, despite near-identical agreement statistics. We therefore do not claim CRUCIBLE eliminates the need for human oversight; we show instead that periodic, targeted human validation—at the scale used here, a small stratified sample per domain—is advisable, and appears necessary before treating automated agreement alone as evidence of validity in settings similar to this one, to determine *which* domains’ automated convergence can be trusted, a determination the convergence metric alone cannot make.

We did not recalibrate CRUCIBLE’s rubrics toward the human labels obtained here. Doing so would reintroduce exactly the circularity this validation study is designed to break: a rubric tuned to match a specific human sample provides no evidence of generalization beyond that sample. The role of this validation is diagnostic, not corrective.

### 5.6.4 Ethics Statement (Human Subjects)

The human validation study involved data collection from human participants via the Prolific platform. No formal institutional ethics review or exemption was sought for this study; participation was governed solely by Prolific’s standard platform consent and terms. Participants were compensated at \$12.69/hour, above Prolific’s platform minimum, for a task with a 140-minute estimated completion window. Participation was voluntary; no personally identifiable information was collected beyond participants’ Prolific IDs, which are pseudonymous and not linked to real-world identity in our data. Task descriptions accurately represented the nature of the work (judging transcript content quality) with no deceptive framing. Data was collected and analyzed in aggregate; no individual participant responses are reported in a way that could identify a specific participant. One submission (Technology & AI domain) was rejected and replaced during data collection; see Section 5.6’s Design subsection for the confirmed rejection criteria and Appendix F for the resulting sensitivity check.

## 5.7 Does the Frontier Ensemble That Labeled the Classifier Track Human Judgment?

Section 5.6 validates the *local calibration panel* that CRUCIBLE revises rubrics against. It does not by itself say anything about the separate, stronger frontier ensemble (GPT-4o mini, Gemini 2.5 Flash, Llama 3.3 70B) that actually produced the labels used to train and evaluate the downstream SNR classifier (Section 5). We close this gap directly using the same 30-item human reference set, since this frontier ensemble’s per-transcript labels for the original 60-transcript corpus were already available.

The production frontier ensemble that actually generated the SNR classifier’s training and evaluation labels tracks human judgment about as loosely as the local calibration panel does—no individual frontier judge exceeds 60% agreement with human consensus, and the ensemble majority (56.7%) is comparable to the local panel’s own 63.3% (Table 4). This is not a minor footnote-level caveat: it is direct evidence that the downstream classifier’s  $F1 = 0.871$  result (Section 5, reported there as a usability case study, not a claim about agreement with the available human reference) rests on labels of similar human-alignment reliability to the ones scrutinized throughout this paper, not on some independently more trustworthy labeling process. Local-panel and frontier-ensemble

Table 5: Frontier ensemble vs. human majority consensus ( $n = 30$ , 10/domain). Gemini 2.5 Flash returned a valid label on 20/30 items; majority vote is robust to this since 2 of 3 judges suffice.

| Comparison                                    | Agreement | Cohen’s $\kappa$ | Gwet’s AC1 |
|-----------------------------------------------|-----------|------------------|------------|
| Frontier ensemble majority vs. human          | 56.7%     | 0.133            | 0.178      |
| GPT-4o mini vs. human                         | 56.7%     | 0.184            | 0.157      |
| Gemini 2.5 Flash vs. human ( $n = 20$ )       | 55.0%     | 0.043            | 0.153      |
| Llama 3.3 70B (Groq) vs. human                | 60.0%     | 0.174            | 0.253      |
| Local calibration panel vs. frontier ensemble | 66.7%     | 0.333            | 0.400      |

labels agree with *each other* more (66.7%) than either agrees with humans—consistent with this paper’s broader finding that automated agreement, even between differently-sourced judge ensembles, does not imply agreement with the available human reference.

**Missing-label sensitivity check.** Gemini 2.5 Flash failed to return a valid label on 10 of the 30 human-validation items; these 10 items are excluded from Gemini’s own individual-agreement row in Table 5 only, and the ensemble-majority row above uses whichever 2 or 3 of the 3 judges returned a valid label on each item (majority requires only 2 of 3, so this is well-defined throughout). On the 10 items where Gemini’s label is missing, the remaining two judges (GPT-4o mini and Llama 3.3 70B) agree with each other on 9 of 10 and disagree on 1 of 10, so Gemini’s absence rarely leaves the majority ambiguous. As a direct check, restricting to the 20 items where all three frontier judges returned a valid label (dropping the 2-of-3 convenience entirely) gives frontier-majority-vs-human agreement of 55.0% (Cohen’s  $\kappa = 0.100$ , Gwet’s AC1 = 0.120)—close to the full-sample, 2-of-3-robust figure of 56.7% ( $\kappa = 0.133$ , AC1 = 0.178) reported above, so the missing-label handling does not materially change the finding.

## 5.8 Rubric Portability Across Judge Models

A natural question is why a single frontier model, applied zero-shot, would not suffice in place of the full calibration loop. We test this directly, and in doing so uncover a second finding: the calibrated rubric does not transfer cleanly to a judge model different from the one it was calibrated on.

### 5.8.1 Frontier Baseline Design

We compare three approaches against the same 30-transcript human ground truth used in Section 5.6: (1) the local three-judge panel applying its calibrated rubric (the system described in this paper); (2) GPT-5.6 Sol [23] (OpenAI, general availability July 9, 2026) applying a naive zero-shot prompt with no rubric; and (3) GPT-5.6 Sol applying the same calibrated rubric used by the local panel. All GPT-5.6 Sol calls for this experiment were made on July 19, 2026, ten days after general availability. We use a non-Anthropic model for this baseline because Claude serves exclusively as the calibration coordinator throughout this work and was not evaluated as a judge, to avoid conflating baseline performance with the model family responsible for rubric design. GPT-5.6 Sol does not support temperature=0 at time of writing; results reflect the model’s default sampling temperature.

### 5.8.2 Findings

Overall, the local calibrated panel and a single zero-shot frontier pass achieve identical accuracy against human judgment (63.3%)—the 10-iteration calibration loop does not categorically outper-

Table 6: Accuracy against independent human judgment (n=30, 10 per domain), comparing the local calibrated panel against a frontier model with and without the calibrated rubric.

| Approach                         | Career     | Tech & AI | Gen. Ed.    | Overall      |
|----------------------------------|------------|-----------|-------------|--------------|
| Local panel + calibrated rubric  | 50%        | 50%       | 90%         | <b>63.3%</b> |
| GPT-5.6 Sol, no rubric           | 60%        | 30%       | <b>100%</b> | <b>63.3%</b> |
| GPT-5.6 Sol, + calibrated rubric | <b>20%</b> | 40%       | 90%         | 50.0%        |

form a single frontier zero-shot pass in aggregate. However, giving the frontier model the *same* calibrated rubric text does not add value, and in Career & Self-Improvement, degrades accuracy substantially (60%  $\rightarrow$  20% against human judgment). Comparing the same frontier model’s outputs against the local panel’s labels directly (rather than against human judgment) shows the rubric *does* make GPT-5.6 Sol’s outputs converge toward the local panel’s specific behavior (agreement with the local panel rises from 0.500 to 0.633 when the rubric is added). Taken together, this indicates the calibrated rubric teaches a frontier judge to mimic the local ensemble’s particular behavior pattern—including the local ensemble’s specific errors—rather than making the frontier judge more accurate against the available human reference. The rubric appears co-adapted to the specific quirks of the local judge ensemble it was calibrated on (Section 5, which documents Qwen 2.5 7B’s specific misreadings that motivated each revision), and this adaptation does not generalize to a different judge model.

We verified this baseline is not a single-draw artifact: repeating both the no-rubric and rubric-prompted GPT-5.6 Sol calls five times per transcript (600 total calls) shows 87–93% of transcripts receive an identical label across all five draws, and the majority-of-five accuracy against the available human reference matches the original single-draw result exactly for both variants.

### 5.8.3 Reasoning-Before-Label Ablation

Motivated by the question of whether Career & Self-Improvement’s stagnation (Table 1) reflects a single-pass prompting limitation rather than a true model capability ceiling, we added an intermediate reasoning step to the local judges’ prompts, in the style of chain-of-thought prompting [20] (reasoning generated before the label, not after), and re-scored all three domains against their existing calibrated rubrics, using the same three local models at temperature=0.0. We call this a *reasoning-before-label prompt* rather than chain-of-thought throughout, since we have no access to the models’ actual internal computation—only to text generated before the label in response to an instruction to reason first.

Table 7: Reasoning-before-label ablation across all three domains: inter-judge weighted agreement and accuracy against the available human reference, without vs. with the reasoning step.

| Domain              | Inter-judge A |              | vs. Human    |              |
|---------------------|---------------|--------------|--------------|--------------|
|                     | No reasoning  | Reasoning    | No reasoning | Reasoning    |
| Career & Self-Impr. | 0.684         | <b>0.767</b> | 0.500        | <b>0.400</b> |
| Technology & AI     | 0.817         | <b>0.933</b> | 0.500        | 0.500        |
| General Education   | 0.967         | 0.917        | 0.900        | 0.900        |

The reasoning step reliably raises inter-judge agreement in the two domains that had room to improve (Career: +8.3 points weighted; Technology & AI: +11.6 points, Gwet’s AC1 nearly triples), but it never *helps*, and in Career actively *hurts*, agreement with human judgment. Career’s human-

agreement drops from 50% to 40% with the reasoning step (Cohen’s  $\kappa$  flips from +0.14 to −0.25); Technology & AI is flat on raw agreement but its AC1 drops; General Education, already near-saturated, is unchanged against humans and slightly dips on inter-judge agreement, consistent with a ceiling effect.

This result rules out simple single-pass prompting failure as the sole explanation for Career & Self-Improvement’s stagnation—the judges *can* be made to agree more with reasoning added—but shows that forcing reasoning-before-label makes the panel more internally consistent without making that consistency more correct. We revise our diagnosis accordingly: the original stagnation reflects not an inability of the judges to converge, but a shared misalignment between judge and human notions of “substantive” career content, one that explicit reasoning reinforces rather than resolves. We used a single, untuned reasoning-before-label prompt design across all domains; whether a more carefully engineered prompt would alter this pattern is untested and left for future work.

### 5.8.4 Revision-Sensitivity Analysis

The findings above raise a further question: how much of the coordinator’s v1 revision’s effect (Technology & AI, 76.7% → 81.7%) is attributable to the coordinator’s specific diagnostic reasoning, versus simply having *any* reasonable revision informed by the disagreement evidence, versus having any revision at all? We call this a *revision-sensitivity analysis*, not an ablation: a single alternative revision per condition cannot isolate the coordinator’s causal contribution, only indicate whether the pattern is sensitive to how the revision is produced.

We regenerated the v0-iteration disagreement cases for Technology & AI’s 20 transcripts via a fresh deterministic pass (temperature=0.0; these were not persisted from the original run, so this is how they are recovered) and generated two alternative revisions, both via GPT-5.6 Sol<sup>4</sup>: (a) *disagreement-informed*, seeing the same disagreement cases and criterion quotes the original coordinator saw, but not the coordinator’s own revision text; and (b) *rubric-only*, seeing only the v0 rubric with no disagreement evidence at all. Both alternatives were run through the same local three-judge panel on the same 20 transcripts, single inference pass, no further iteration.

Table 8: Revision-sensitivity analysis, Technology & AI (n=20). Regenerated v0 matches the originally reported 76.7% exactly, indicating no detectable model drift since the original run.

| Condition                                   | Weighted $A$ | Fleiss $\kappa$ | Gwet AC1     |
|---------------------------------------------|--------------|-----------------|--------------|
| v0 baseline (regenerated)                   | 0.767        | 0.067           | 0.067        |
| Original coordinator v1 (Claude Sonnet)     | <b>0.817</b> | <b>0.246</b>    | <b>0.287</b> |
| Alt (a) disagreement-informed (GPT-5.6 Sol) | 0.734        | −0.111          | −0.026       |
| Alt (b) rubric-only (GPT-5.6 Sol)           | 0.767        | 0.063           | 0.071        |

In this single trial, the original coordinator outperformed both alternatives, and having disagreement evidence was not by itself sufficient: alt (a), given the *same* evidence the coordinator saw, performed *worse* than alt (b)’s blind, no-evidence revision, which in turn was statistically indistinguishable from not revising at all. Looking at the generated text, alt (a) rewrote multiple HIGH criteria broadly and more restrictively, where the original coordinator made a single surgical clause edit (Appendix A); the broad rewrite appears to have destabilized rather than fixed the panel. We

<sup>4</sup>No Anthropic API access was available for this analysis; both alternatives substitute GPT-5.6 Sol for Claude Sonnet, the original coordinator model. This is a genuine confound: differences from the original coordinator’s v1 rubric may partly reflect model-family differences in rubric-writing style rather than purely the evidence-visibility manipulation this analysis targets.

do not read this as establishing that the coordinator’s diagnostic reasoning is necessary or superior in general: both alternatives were generated by a different model than the original coordinator, so this comparison cannot cleanly separate “the coordinator’s reasoning is better” from “Claude Sonnet writes better surgical rubric edits than GPT-5.6 Sol for this task.” We report it as a sensitivity finding, not a controlled ablation result.

These findings—rubric non-transfer across judge models, a reasoning-before-label step increasing agreement without increasing agreement with the available human reference, the frontier ensemble tracking human judgment about as loosely as the local panel (Section 5.7), and the Section 5.6 finding that Technology & AI’s confident convergence masked a divergence from human judgment—share a common implication demonstrated several ways: **making automated judges agree more with each other is not the same as making them more correct**. Section 6.3 adds a fifth, more direct demonstration of the same point via judge replacement. We note the human-validation, rubric-portability, and reasoning-before-label analyses in this section share a common 30-item human-validation reference set ( $n=30$ ); while each tests a distinct manipulation, they should be read as related diagnostic analyses on one reference set, not as independent replications, since a property specific to this particular set could in principle affect several of them jointly.

## 6 Analysis and Discussion

Every result below is organized around the two questions posed in the Introduction: does disagreement-driven rubric revision generalize past the transcripts it was tuned on (Section 5.3), and does higher inter-judge agreement correspond to higher human alignment (Sections 5.6, 5.7, 5.8, and 6.3 below). Table 9 summarizes when each analysis was run relative to the frozen system, to make explicit that nothing downstream of the primary calibration run fed back into rubric or system design.

Table 9: Chronology of analyses. “Could influence design?” marks whether a stage’s output could, in principle, have fed back into rubric or system design. Only the first two stages could; every analysis from human validation onward was run against the frozen system and rubrics.

| Stage                                   | Data used                                    | Could influence design? |
|-----------------------------------------|----------------------------------------------|-------------------------|
| Initial rubric design (v0)              | Domain knowledge, no corpus data             | Yes                     |
| Primary calibration run                 | 60-transcript corpus, iter. 1–10             | Yes                     |
| Human validation (§5.6)                 | 30-item subset (10/domain)                   | No                      |
| Frontier ensemble vs. human (§5.7)      | Same 30-item subset                          | No                      |
| Rubric portability (§5.8)               | Same 30-item subset                          | No                      |
| Reasoning-before-label (§5.8.3)         | Same 30-item subset                          | No                      |
| Revision-sensitivity (§5.8.4)           | Tech & AI subset (alt. rubrics, not adopted) | No                      |
| Held-out rubric evaluation (§5.3)       | 90-transcript unseen-to-coordinator set      | No                      |
| Judge replacement (§6.3)                | Career subset                                | No                      |
| Irreducible-case sensitivity (App. E)   | Existing calibration output                  | No                      |
| Prolific rejection sensitivity (App. F) | Existing annotation records                  | No                      |

### 6.1 What CRUCIBLE Addresses

CRUCIBLE substitutes structured inter-judge disagreement—measurable, diagnosable, and actionable—for the implicit and unauditable disagreement that characterizes single-annotator labeling.

This is not a claim that ensemble labels are unbiased. Each judge carries its own training biases.



But the ensemble structure makes those biases visible: when one model systematically diverges from the other two, the coordinator can name the pattern. Human annotator bias is usually invisible unless multiple annotators are compared—precisely the expensive intervention this system replaces.

## 6.2 The Capability Ceiling Finding

The Career & Self-Improvement ceiling—a best of 70.0%, short of convergence—shows that a class of disagreement exists that rubric revision did not fix under this judge and prompt setup. Without the CRUCIBLE diagnostic loop, an operator seeing 70.0% would not know whether to keep revising or replace the judge. The stagnation pattern plus the coordinator’s consistent diagnosis across both revision attempts—attributing it to a model-level inability to track co-occurrence patterns across long documents—suggested the rubric was not the limiting factor. The natural prescription is judge replacement or ensemble augmentation, not further revision; Section 6.3 tests this directly, though as that section shows in detail, the specific test we were able to run cannot fully distinguish a genuine judge-level capability limitation from a structural property of this particular three-judge panel.

## 6.3 Judge Replacement: A Direct Test

Section 6.2 recommends judge replacement, not further rubric revision, once a domain is diagnosed as capability-bounded. We test this directly on Career & Self-Improvement, replacing Qwen 2.5 7B—the judge the coordinator’s diagnosis consistently implicated—with **granite3.1-dense:8b** (IBM, 8.2B parameters, 131,072-token context), keeping Llama 3.1 8B and Mistral 7B unchanged and the retained v1 rubric fixed. This is the only replacement model attempted: we selected it before running any evaluation, for comparable scale to Qwen 2.5 7B and a training lineage distinct from Meta, Mistral AI, and Alibaba—explicitly not Gemma 2 9B, which Section 3.5 already disqualified as degenerate. Trying multiple candidates and reporting the best-performing one would constitute model selection on the same small human-reference set used to score it below, which we specifically avoided.

Table 10: Judge replacement on Career & Self-Improvement (n = 20 for inter-judge agreement; n = 10 for human alignment).

| Panel                                 | Inter-judge A | vs. Human | vs. Human ( $\kappa$ / AC1) |
|---------------------------------------|---------------|-----------|-----------------------------|
| Original (Llama, Mistral, Qwen)       | 0.700         | 50.0%     | 0.138 / 0.083               |
| Replacement (Llama, Mistral, Granite) | <b>0.883</b>  | 50.0%     | 0.138 / 0.083               |

Replacing the diagnosed-bottleneck judge raises inter-judge agreement substantially, from 0.700 to 0.883. Accuracy against the 10 human-labeled Career transcripts is *bit-for-bit identical* before and after: the replacement panel’s majority-vote label matches the original panel’s on every one of the 10 items, so agreement, Cohen’s  $\kappa$ , and Gwet’s AC1 against human judgment are unchanged to three decimal places.

**Why: a mechanical check on the full 20-transcript set.** We looked more closely at why, using the full 20-transcript Career set (not just the 10 with human labels), and found this dissociation is mechanically guaranteed rather than an independently interesting empirical outcome. Llama 3.1 8B and Mistral 7B—the two judges held fixed across both panels—already agree with each other on 18 of the 20 (90%) Career transcripts (Llama predicts HIGH on 20/20, Mistral on 18/20); on those 18, the majority-vote label is determined entirely by Llama and Mistral, regardless of what the

third judge outputs. On the remaining 2 transcripts (Llama = HIGH, Mistral = LOW), both the original judge (Qwen 2.5 7B, 10% HIGH rate on this domain) and the replacement (granite3.1-dense, 65% HIGH rate) happened to vote LOW, so the majority stayed LOW under both panels there too. The result: the majority-vote label changed on **0 of the 20** Career transcripts between the original and replacement panel. The inter-judge agreement metric moved (0.700  $\rightarrow$  0.883) because granite3.1-dense agrees with the Llama+Mistral majority far more often than Qwen did (65% vs. 10% HIGH rate), converting many 2-1 splits into 3-0 unanimity—not because the panel’s actual output changed on any transcript.

This means the finding is narrower than the table alone suggests: what we have directly demonstrated is that judge replacement changed whether the panel was unanimous or split, not what the panel’s actual annotations were, and by construction the annotations *could not* have changed on this particular panel composition. This result is *consistent with*—but does not independently establish—this paper’s broader claim that fixing inter-judge agreement does not fix agreement with human judgment: the majority-vote label here was mechanically unchangeable by any third judge given how strongly the other two already agreed, so the null result on human alignment cannot be attributed with confidence to *correcting* a genuine capability gap versus simply *being unable to move* the panel’s output at all. The label prevalence underlying this is informative too: both panels produce the same distribution of HIGH/LOW calls at the majority-vote level on this subset, which is consistent with the two panels reaching the same place by different within-panel routes rather than the replacement judge correcting Qwen’s specific misreadings—though, per the analysis above, “reaching the same place” was close to inevitable given the fixed pair’s own agreement rate.

We read this cautiously in two respects: it is a single replacement-model trial on one domain (see Limitations), and, as detailed above, the extent to which any third judge could have influenced the majority-vote outcome at all is itself limited on this panel. This experiment is better read as evidence that judge replacement can convert a split panel into a unanimous one without changing its output, rather than as a demonstration that judge replacement is a general recovery strategy for capability-bounded disagreement (see Limitations for a fuller discussion of this constraint).

## 6.4 Degenerate Judge Detection

A judge assigning a single label to all documents makes agreement appear high when it is driven by shared degeneracy rather than shared understanding. We treat discriminative validity as a pre-screening gate: no model enters a judge panel without demonstrating a non-degenerate label distribution across a diagnostic sample.

## 6.5 Output-Stability Recheck for Converged Domains

Once a domain converges, we re-evaluate it with the same rubric and the same three judge models on the iteration immediately following convergence, checking whether the weighted agreement rate holds. Each coordinator call is a fresh, stateless API request scoped to a single domain (it receives only that domain’s rubric, disagreement cases, and revision history; no information about other domains’ prompts, revisions, or outputs is ever included), so this check is not, and cannot be, a defense against one domain’s revision “priming” the coordinator’s phrasing on another domain via shared state—no such state exists in this implementation. What it does test is judge-level output stability: whether the same rubric applied to the same transcripts by the same three models reproduces the same agreement rate on a second pass, since a frontier API coordinator’s own non-determinism (Limitations) is a property of the coordinator, not the judges, but serving-side

variance in the local judge models is not fully ruled out either. No regression (agreement drop  $> 10$  percentage points) was observed in the run reported here across any of the domains checked.

## 7 Limitations

**Single experiment.** Results are for 60 transcripts across three domains. Generalization requires validation on other domains and document types.

**Model-specific ceiling.** The Career stagnation was diagnosed as specific to Qwen 2.5 7B under our prompt and setup; we cannot rule out that a different prompt formulation or judge model would shift the ceiling. Future work should evaluate judge replacement and prompt variation as recovery strategies. This attribution rests on a single, untuned prompt template per judge model used throughout the calibration run reported here; prompt reformulation remains untested as an alternative explanation for the ceiling and should be evaluated alongside judge replacement before concluding the ceiling is inherent to the model rather than an artifact of this specific prompt.

**Judge replacement: tested once, narrowly.** Section 6.3 now tests Section 6.2’s recommendation directly: replacing Qwen 2.5 7B with granite3.1-dense:8b on Career & Self-Improvement raised inter-judge agreement substantially ( $0.700 \rightarrow 0.883$ ) while leaving accuracy against the 10 human-labeled transcripts completely unchanged. This is a single replacement-model trial on a single domain, with only one candidate model attempted; it does not establish that judge replacement is a general recovery strategy.

**Judge replacement’s strength depends on how much the third judge could actually influence the outcome.** As Section 6.3 details, the two retained judges (Llama, Mistral) already agreed with each other on 90% of this domain’s 20 transcripts, and on the remaining 2 both candidate third judges happened to vote the same way; the majority-vote label therefore changed on 0 of 20 transcripts between the original and replacement panel. This means the experiment, as run, could only ever have shown a dissociation between inter-judge agreement and human alignment on this domain—it was structurally incapable of showing the opposite, because the panel’s output could not move. A domain or panel where the two retained judges disagree with each other more often (so the third judge’s identity genuinely determines more majority votes) would be a stronger, more informative test of whether judge replacement recovers human alignment along with inter-judge agreement; we did not have such a case available in this run’s data. Readers should weight this result accordingly: it demonstrates a real property of the diagnostic loop (replacement resolved the agreement symptom) but the null result on human alignment is consistent with, rather than independent confirmation of, this paper’s broader claim.

**Silver labels.** Calibrated labels retain the biases of the three judge models. Section 5.6’s human validation study addresses part of this—human majority consensus matches CRUCIBLE’s calibrated label 90% of the time in General Education but only 50% in Technology & AI and Career & Self-Improvement—but that study itself uses a small ( $n = 10/\text{domain}$ ) unaided-judgment human sample (see below), not a large-scale or rubric-guided gold standard. For high-stakes tasks, human adjudication of remaining disagreements at greater scale is recommended.

**Synthetic data in classifier training.** Of the 389 transcripts used to train the downstream SVM classifier (Section 5), 300 (77%) are synthetically generated with labels assigned by construction, intended to match the generating prompt’s target class, rather than assigned by the calibrated-rubric ensemble. The reported  $F1 = 0.871$  may therefore partly reflect the classifier separating synthetic-generation artifacts from real transcripts, rather than purely the quality of the calibrated

real-transcript labels. A real-transcript-only training ablation would be needed to isolate how much of the reported F1 is attributable to the calibrated labels specifically.

**Coordinator dependency.** Revision quality depends on the coordinator model. A weaker coordinator may misdiagnose disagreement or introduce new ambiguities. Retry logic mitigates transient failures but not systematic ones. Because the coordinator is a frontier API model and is not bit-reproducible across runs, we expect the exact per-domain ceiling to vary by roughly one to two points between runs, based on typical run-to-run variance for this kind of model; we have not verified this band empirically, since, as noted immediately below, only a single calibration run was performed. We accordingly do *not* claim the qualitative findings (the capability ceiling, degenerate-judge detection, and the three agreement regimes) are empirically robust to run-to-run variation—only that they are unlikely to be artifacts of a single anomalous coordinator call, given that each rests on a consistent diagnosis repeated across multiple iterations within the one run we did perform.

**Single calibration run.** The reported convergence trajectory (Table 2) reflects a single run of a non-deterministic coordinator, and no repeated-run robustness check was performed. In particular, Career & Self-Improvement’s iteration-2 gain (68.4%  $\rightarrow$  70.0%, 1.6 points) falls within the estimated  $\pm 1$ –2 point run-to-run variance band described above; that specific incremental result should therefore be read cautiously as a possible artifact of coordinator noise rather than confirmed as a genuine improvement from the rubric revision. Repeated runs would be needed to establish it as signal rather than noise, and more generally to establish empirically (rather than argue plausibly) that this paper’s qualitative findings are robust to coordinator variation.

**No baseline comparison, partially addressed.** We did not benchmark CRUCIBLE against manual rubric revision or simple self-refinement; these remain future work. A single frontier judge *is* now compared, both without a rubric and with the calibrated rubric (Section 5.8), so that specific gap is closed. Isolating the coordinator-mediated loop’s contribution from these still-untested baselines remains open.

**Coordinator contribution: sensitivity-tested, not isolated.** Section 5.8.4 now compares the original coordinator’s revision against two alternatives on Technology & AI—one given the same disagreement evidence, one given only the rubric—and finds the original coordinator outperformed both, while the disagreement-informed alternative underperformed the blind one. This is informative but does not isolate the coordinator’s causal contribution: both alternatives were generated by a different model (GPT-5.6 Sol, substituting for the unavailable Claude Sonnet coordinator), so the comparison cannot cleanly separate “the coordinator’s diagnostic reasoning is better” from “Claude Sonnet writes better surgical rubric edits than GPT-5.6 Sol for this task,” and only one alternative per condition was tried on one domain. A same-model ablation comparing coordinator-proposed revisions against alternative revisions of similar scope—ideally using Claude Sonnet for all conditions, or adding human-written alternatives—would more cleanly isolate whether the coordinator’s specific diagnosis matters. This remains future work.

**Small human-validation sample.** Our human validation (Section 5.6) uses  $n = 10$  real transcripts per domain, plus one attention check, a scale chosen as a practical minimum sufficient for this purpose. With  $n = 10$  per domain, the percentile bootstrap intervals in Table 4 may not fully capture the true sampling uncertainty and should be read as indicative rather than precise; they are also wide (Technology & AI’s and Career & Self-Improvement’s overlap substantially), and our domain-level distinction rests on the joint pattern of human-human and human-model agreement, not on point estimates alone. Channel identity is recorded for this corpus, and resampling by

channel rather than by transcript (Appendix G) changes these intervals only slightly (overall upper bound 80.0%  $\rightarrow$  79.3%), since most channels contribute only one transcript to this particular 30-item sample—but the underlying 60-transcript corpus does have repeated channels (up to 5 transcripts from one channel), so this should not be read as ruling out channel-level dependence in general, only as showing it is small for this specific sample. A larger-scale replication—human evaluation across the full 60-transcript corpus—would narrow these intervals and is scoped as future work. Relatedly, for Career & Self-Improvement specifically (human AC1 = 0.33), the 2-of-3 majority vote used as “human consensus” throughout Sections 5.6, 5.7, and 5.8 is itself a noisy estimate given how weakly the three annotators agree with each other on this domain; annotator-level uncertainty is not propagated into the human-vs-CRUCIBLE comparison (Table 4) or the frontier-baseline comparison (Table 6), both of which treat the majority label as a fixed point estimate. Separately, our human task uses an unaided-judgment design only (Section 5.6’s Design subsection): annotators were never shown the calibrated rubric. This tests whether CRUCIBLE’s labels align with unaided human intuition, not whether humans agree with the rubric’s specific operational criteria when shown it; a two-condition design (unaided plus rubric-guided) would additionally test construct validity of the rubric’s criteria themselves and is scoped as future work.

**Held-out generalization is narrower than it may sound.** Section 5.3’s positive result generalizes the rubric revision’s effect on *inter-judge agreement*, not on agreement with a human reference (no human labels exist for that 90-transcript corpus), and the “unseen-to-coordinator” claim rests on documented project workflow, not a logged, timestamped audit trail ruling out the corpus’s exposure during development. The two corpora also share 6 channels (Section 4), so this is document-held-out, not source-held-out. Most materially, the effect is driven disproportionately by one judge (Qwen 2.5 7B) shifting its label distribution toward the other two, largely static, judges—not by uniform three-way convergence toward more nuanced shared judgment—and the underlying majority-vote annotation changes on **0 of the 90** held-out transcripts between the v0 and final rubric (Section 5.3), so the generalization result should not be read as evidence the calibrated rubric produces more broadly reliable judgment on unseen data, or that it changes any actual output label on this set, only that it reliably shifts one judge’s individual predictions in a direction that happens not to move the already-fixed majority.

**Frontier baseline sampling.** GPT-5.6 Sol does not expose temperature control; our stability check (Section 5.8) characterizes label stability under the model’s default sampling rather than variance at a fixed, controlled temperature, and cannot separate genuine sampling noise from serving-side non-determinism.

**Reasoning-before-label scope.** The reasoning-before-label ablation (Section 5.8.3) uses a single, untuned prompt design tested across all three domains but not across alternative formulations; whether more carefully engineered reasoning prompts alter the pattern of raised inter-judge agreement without raised agreement with the human reference is untested.

**Shared evaluation reference.** The human validation, frontier-vs-human comparison, rubric-portability, reasoning-before-label, and the human-alignment portion of the judge-replacement analysis (Sections 5.6, 5.7, 5.8, 5.8.3, and 6.3) are all evaluated against the same 30-transcript human-validation set (or, for judge replacement, its Career subset). They test distinct manipulations but are related diagnostic analyses on one reference set, not independent replications in the statistical sense: a property specific to this particular reference set could in principle affect several of them jointly. The revision-sensitivity analysis (Section 5.8.4) and held-out evaluation (Section 5.3) use different corpora and are not subject to this specific concern, though each has its own scope limits noted above.

**Reproducibility of the original run’s exact configuration.** Before treating anything as unrecoverable, we checked available sources for the primary calibration run’s exact configuration. Recovered: the run’s timestamp (2026-05-31, from `calibration_summary.json` and corroborated by output file modification times); current digests, quantization (Q4\_K\_M for all three original judges), and context window (32,768 tokens for Qwen 2.5 7B and Mistral 7B, 131,072 for Llama 3.1 8B) via `ollama show`, which local file modification timestamps (“7 weeks” prior to this writing) are consistent with, but do not prove, matching what was used in the original run; and confirmation from the judge-calling code itself (`judge_agent.py`) that temperature was explicitly set to 0.0 and no context-window override was applied, so each model’s default context window applied throughout—ample for this task’s prompts (transcripts capped at 3,000 characters plus a short rubric, well under any of these limits), which rules out context-window truncation as an explanation for Qwen’s specific misreadings, addressing one previously open question. Not recoverable: shell history containing the original `ollama pull` commands and their exact timestamps (none found); API billing or usage logs for the Claude Sonnet coordinator calls (not accessible in this environment); and, most importantly, a saved digest snapshot from *the original run itself* to diff against today’s digests, since none was captured at the time. Follow-up experiments in this paper (Sections 5.3, 6.3, and 5.8.4) were conducted under this closest recoverable configuration and should be interpreted as diagnostic extensions rather than exact controlled replications of the original run.

**Runtime.** At 17–20 minutes per iteration, a 10-iteration run requires nearly three hours. Parallelizing judge evaluation across models would reduce iteration time by roughly two-thirds.

## 8 Ethics Statement

This work’s human validation component (Section 5.6) involved data collection from human participants via Prolific. No formal institutional ethics review or exemption was sought; participation was governed solely by Prolific’s standard platform consent and terms. Participants were compensated at \$12.69/hour, above the platform minimum, for tasks with a 140-minute estimated completion window. No personally identifiable information beyond pseudonymous Prolific IDs was collected. Task descriptions accurately represented the work with no deceptive framing. All other experiments in this paper (rubric calibration, frontier baseline comparisons, the reasoning-before-label ablation, held-out evaluation, judge replacement, and revision-sensitivity analysis) involve no human subjects and use publicly available or synthetically generated transcript data.

**Use of generative AI.** Generative AI tools were used in this work for code generation and debugging of the experimental pipeline, and for writing refinement and editorial revision of the manuscript. All experimental design decisions, result interpretation, and scientific claims are the author’s own; all reported numbers derive from the released code and data, and the author reviewed and takes full responsibility for the final content.

## 9 Conclusion

CRUCIBLE demonstrates that annotation rubric calibration can be automated, after an initial rubric draft, through a coordinator-mediated iterative loop driven by inter-judge agreement signal, with no human reference labels used during optimization and no human in the loop after initialization (every retention, rollback, irreducibility-flagging, and stopping decision in the loop is programmatic; see Section 3.4). The system was built to solve a concrete problem—an earlier, differently-scoped pilot showed inconsistent, largely-uncalibrated judge disagreement (Introduction, Appendix B)—and raw weighted agreement using the local calibration panel reached 81.7%–96.7%



on two domains, though chance-corrected agreement shows the genuine gain concentrates in Technology & AI. This paper’s central finding, however, is not that number: it is that raising or achieving inter-judge agreement is not the same as achieving agreement with human judgment, demonstrated five ways (Sections 5.6, 5.7, 5.8, 5.8.3, and 6.3)—most directly by judge replacement, which raised inter-judge agreement from 0.700 to 0.883 while leaving human agreement bit-for-bit unchanged. As Section 6.3 details, that specific result is narrower than it first appears: the two retained judges already agreed with each other on 90% of the domain’s transcripts, so the panel’s majority-vote output was mechanically unchangeable by any third judge, and the null result on human alignment is consistent with, rather than independent proof of, the broader claim. The framework’s durable value is its diagnosis of disagreement structure, a check that this diagnosis’s effect on the inter-judge *agreement metric* generalizes past the calibration corpus even though the underlying majority-vote annotations do not change on the held-out set (Section 5.3), and the downstream classifier case study (F1=0.871) the calibrated rubrics enabled.

Beyond the specific numbers, CRUCIBLE surfaces one finding with broader implications: the capability ceiling, and its limits. Stagnation under coordinator-mediated revision is a diagnostic—it identifies that the limiting factor is plausibly related to a judge’s model-level capability rather than rubric underspecification, though our judge-replacement test cannot cleanly separate a genuine capability limitation from a structural property of a panel where two judges already agree most of the time, and motivates judge replacement or ensemble augmentation as directions worth testing over further revision. Section 6.3 tests judge replacement directly and shows it resolves the agreement symptom completely without changing the panel’s output at all, on the one domain and one replacement model tested, and by construction could not have changed the output on this specific panel—a result that sharpens rather than closes the diagnostic question. On a practical note, the columnar evaluation order (batching all transcripts per model before switching) reduced per-iteration model-load events from 180 to 3, making a full 10-iteration run tractable on an Apple Silicon MacBook with no discrete GPU (exact CPU/GPU offload configuration for the Ollama-served models was not recorded).

Code, calibration logs, versioned rubrics, and calibrated labels are publicly available at <https://github.com/biditdas18/rubric-calibration-agent>.

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## A Rubric Calibration: Before and After

We show the initial (v0) and final calibrated rubric text for each domain. Only HIGH-signal criteria are shown for brevity; full rubrics (including LOW-signal criteria and conflict rules) are available in the released `calibrated_rubrics.json`.

**General Education (v0 → v0, unchanged).** General Education converged on iteration 1 with zero rubric changes; the initial rubric required no revision.

**Technology & AI (v0 → v1).** *v0*: “Names specific technologies, algorithms, libraries, or architectural patterns with accurate technical detail (e.g. ‘use Redis for session caching’, ‘transformer attention is  $O(n^2)$  in sequence length’)”. *v1 (retained)*: “Names specific technologies, algorithms, data structures, or architectural patterns with accurate technical detail (e.g. stack-based  $O(n)$  solution, Project Mainline module updates, M5 neural engine)”. The revision broadened “libraries” to “data structures” and replaced software-engineering-specific examples with domain-neutral technical examples, addressing Qwen 2.5 7B’s overly narrow reading of the original criterion; this single change raised agreement from 76.7% to 81.7%.

**Career & Self-Improvement (v0 → v1 retained, v2 discarded).** *v0*: “Contains at least 2 specific procedural steps with named methods, tools, or timelines (e.g. ‘File Form I-539 within 60 days’, ‘Use the STAR method for interviews’)”. *v1 (retained as final)*: “Contains at least 2 specific procedural steps with named methods, tools, timelines, or defined techniques—conceptual definitions alone do not qualify but named practices or described processes do”. This revision added an explicit exclusion clause to address Qwen 2.5 7B crediting abstract restatement as procedural content, raising weighted agreement from 68.4% to 70.0%—the domain’s best-observed result. A second revision (v2, not reproduced here as it was not retained) further tightened the co-occurrence condition between motivational language and procedural content; this regressed agreement to 68.4% and was discarded in favor of v1. The revisions shown above increase specificity rather than simplify toward keyword-matching, addressing the concern that rubric calibration might optimize for consensus by trivializing criteria rather than sharpening them.

## B 3-Category Pilot Baseline: Full Figures

During development, an earlier, differently-scoped pilot annotation pipeline used a domain-agnostic rubric applied across a frontier chat-model ensemble (GPT-4, Gemini, Meta AI), reported in the companion SNR paper [24], Section 4.1. That pilot used a 3-category label space (HIGH/MID/LOW) later collapsed to binary via majority vote for downstream use, computed from the raw per-judge labels: 41.1% average pairwise agreement across the three judge pairs, and 16.7%

full three-way unanimity (10/60 transcripts fully unanimous; 44/60 with exactly one matching pair; 6/60 with all three judges giving different labels). The 6/60 fully-split rate is possible because the label space has 3 categories, not 2 as in CRUCIBLE’s own metric—we do not attempt to reconcile the 41.1% and 16.7% figures via a binary-case formula, since that formula does not apply to a 3-category space; both figures are independently correct as computed directly from the raw per-judge labels.

**Non-comparability.** This pilot (3-category label space, a different model ensemble, pairwise/unanimity metrics) and CRUCIBLE’s own calibration metric (binary label space, the local judge panel, the weighted- $A$  metric of Eq. 1) are measured in different label spaces with different models and are **not** directly comparable in magnitude. We present the pilot as qualitative motivating context—an independently-run experiment showing low, inconsistent agreement in an uncalibrated, differently-scoped setup—not as a quantitative pre/post baseline for CRUCIBLE’s own reported convergence numbers (76.7%–96.7% weighted  $A$ ). No percentage-point improvement between the pilot and CRUCIBLE’s results is computed or implied anywhere in this paper.

## C Columnar Evaluation Order: Full Derivation

The naive *row-wise* approach:

$$T_1 \rightarrow [\text{Llama, Mistral, Qwen}] \rightarrow T_2 \rightarrow [\text{Llama, Mistral, Qwen}] \rightarrow \dots$$

requires Ollama to load, run, and unload three model weight sets for every transcript. On consumer hardware with limited VRAM, each model switch incurs a cold-load penalty. Across 60 transcripts and 3 models per transcript, this produces 180 model-load events per iteration. In practice, this scheme could not complete a single iteration within 30 minutes and was cancelled.

The fix draws on a principle from columnar database design: group operations by column (model) rather than by row (transcript):

$$\underbrace{\text{Llama} \rightarrow 60 \text{ transcripts}}_{\text{col. 1}} \rightarrow \underbrace{\text{Mistral} \rightarrow 60 \text{ transcripts}}_{\text{col. 2}} \rightarrow \underbrace{\text{Qwen} \rightarrow 60 \text{ transcripts}}_{\text{col. 3}}$$

This reduces model-load events from 180 to 3 total per iteration. Each model loads once and remains resident in memory for its full 60-transcript pass. Result: each iteration completed in 17–20 minutes on a consumer MacBook (Apple Silicon, no discrete GPU; exact CPU/GPU offload configuration not recorded), making a 10-iteration calibration run (170.7 minutes total; the run terminated at the maximum-iteration cap of 10 iterations) fully tractable.

## D Held-Out Rubric Evaluation: Full Detail

Full per-domain metrics and per-judge label prevalence (HIGH rate) underlying Table 3 (Section 5.3):

| Domain         | Cond. | All-agree | Pairwise | $\kappa$ | AC1    | Prevalence (L/M/Q) |
|----------------|-------|-----------|----------|----------|--------|--------------------|
| Career & SI    | v0    | 16.7%     | 44.4%    | −0.385   | +0.072 | 100% / 100% / 17%  |
| Career & SI    | Final | 36.7%     | 57.8%    | −0.268   | +0.367 | 100% / 100% / 37%  |
| Tech & AI      | v0    | 53.3%     | 68.9%    | −0.184   | +0.578 | 100% / 100% / 53%  |
| Tech & AI      | Final | 63.3%     | 75.6%    | −0.139   | +0.689 | 100% / 100% / 63%  |
| Gen. Education | Both  | 96.7%     | 97.8%    | −0.011   | +0.977 | 100% / 100% / 97%  |

L/M/Q = Llama 3.1 8B / Mistral 7B / Qwen 2.5 7B. Llama and Mistral predict HIGH on essentially every transcript in both revised domains under both rubric versions; Qwen is the only judge whose prevalence shifts, and by an amount that tracks the reported weighted-*A* improvement closely. Judge calls: 90 transcripts  $\times$  2 rubric versions  $\times$  3 judges = 540, single inference pass each, temperature = 0.0, cached and resumable; script and raw per-call cache are released with this paper’s code.

## E Irreducible-Case Audit

**Independent audit: infeasible.** We attempted to independently re-examine the coordinator’s irreducible-case calls (Section 5 originally named two specific Technology & AI transcripts) against the actual per-case criterion quotes the coordinator saw. No such log survives: inspection of `calibration_loop.py` and `coordinator_agent.py` confirms only three files are ever persisted (`calibrated_rubrics.json`, `calibrated_labels.csv`, `calibration_summary.json`); flagged-irreducible transcript indices are tracked only as an in-memory Python set during the run and never written to disk. The specific transcript identities named in an earlier draft of this paper could not be verified against any surviving artifact and have been removed from the main text accordingly.

**Sensitivity check: completed.** Once a transcript is flagged irreducible, `calibration_loop.py` excludes it from that domain’s active evaluation pool for all subsequent iterations (confirmed by code inspection), so it stops contributing to the domain’s tracked weighted-*A* denominator from that point on (the “excluded” treatment, i.e., the number actually reported during the run). We compare this to an “included” treatment: the full re-labeling of all 20 transcripts per domain under the final rubric (`calibrated_labels.csv`), which counts every transcript regardless of irreducible-flag status. This is the most direct recoverable proxy for “included,” not a reconstruction of the original two flagged cases, since their specific identities are not recoverable (above).

| Domain              | Irreducible count | Excluded (reported) | Included (proxy)   |
|---------------------|-------------------|---------------------|--------------------|
| General Education   | 0                 | 96.7%               | 96.7%              |
| Technology & AI     | 2                 | 81.7%               | 81.7%              |
| Career & Self-Impr. | 0                 | 68.4%               | 70.0% <sup>†</sup> |

<sup>†</sup>Career’s difference is *not* an irreducible-case effect (its count is 0, so this check is a structural no-op there); it reflects the pre-existing v1-retained-vs-v2-discarded discrepancy already discussed in Section 5 and Table 2.

For Technology & AI, the domain with irreducible cases, the convergence conclusion (converged, 81.7%, above the 0.80 threshold) is identical under both treatments. As an upper-bound reference, 11 of the 20 transcripts still show inter-judge disagreement in the full final-rubric relabel (global indices 24, 25, 28, 30, 31, 33, 35, 36, 37, 38, 39)—a superset of whichever 2 were originally flagged irreducible, not those cases specifically.

## F Prolific Rejection Sensitivity Check

One of nine human-validation submissions (Technology & AI domain) was rejected and replaced. Cross-referencing the pre- and post-replacement raw batch exports against the officially recorded consensus (`human_validation_ground_truth.csv`) by testing which of the four possible 3-of-4 annotator-response subsets exactly matches the recorded consensus multiset on all 10 non-attention-

check items identifies the rejected annotator unambiguously (10/10 exact match for one subset; 6–7/10 for every other), independently confirmed against the Prolific dashboard’s Rejected tab (single rejected participant, matching the rejected submission’s internal record; completion time 39:06 against this batch’s 140:00 median).

Recomputing this domain’s human consensus with the rejected annotator’s original responses reinstated in place of the replacement annotator’s changes **zero** of the 10 item-level consensus labels. Local-panel-vs-human agreement for Technology & AI is identical under both the official and reinstated consensus (agreement 50.0%, Cohen’s  $\kappa = -0.19$ , Gwet’s AC1 = 0.20, both conditions). The domain-level conclusions reported in Section 5.6 do not depend on this exclusion.

## G Channel-Clustered Confidence Intervals

Full detail underlying the channel-clustering comparison summarized in Limitations. Channel identity for the 60-transcript corpus is available via `video_selection_60.csv` (SNR-Detector repository), joined to this corpus by verified positional alignment. Bootstrap intervals below use 10,000 resamples; “channel-resampled” resamples at the channel level (with all of a sampled channel’s transcripts included together) rather than the transcript level.

| Domain                                    | # channels | Point        | Transcript-CI         | Channel-CI            |
|-------------------------------------------|------------|--------------|-----------------------|-----------------------|
| Career & Self-Impr.                       | 10         | 50.0%        | [20.0%, 80.0%]        | [20.0%, 80.0%]        |
| General Education                         | 9          | 90.0%        | [70.0%, 100.0%]       | [66.7%, 100.0%]       |
| Technology & AI                           | 8          | 50.0%        | [20.0%, 80.0%]        | [22.2%, 77.8%]        |
| <b>Overall (27 unique channels of 30)</b> |            | <b>63.3%</b> | <b>[46.7%, 80.0%]</b> | <b>[46.7%, 79.3%]</b> |

Career & Self-Improvement’s 10 sampled transcripts happen to come from 10 distinct channels, so its channel- and transcript-resampled intervals are identical by construction. General Education and Technology & AI each have one repeated channel within their 10-item sample, producing a small widening or shift at one bound (General Education’s lower bound moves from 70.0% to 66.7%; Technology & AI’s interval shifts from [20.0%, 80.0%] to [22.2%, 77.8%]). The overall effect across all 30 items is small (upper bound 80.0%  $\rightarrow$  79.3%). This should be read as evidence channel clustering has a small effect *on this particular 30-item sample*, not as evidence it would be negligible at different sample sizes or samples with heavier channel repetition, such as the full 60-transcript corpus (up to 5 transcripts from one channel) or the 90-transcript training-supplement corpus (up to 7 from one channel).